

DANIEL FORSYTHE

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SUMMARY

Master of Data Science graduate with hands-on experience building classification and recommendation models in Python, using scikit-learn, TensorFlow and PyTorch, and deploying lightweight APIs with Flask and Docker. Comfortable working across the full pipeline from data cleaning through to model evaluation and presenting findings to non-technical stakeholders. Looking to bring this experience into a Junior Machine Learning Engineer role.

SKILLS

Python, SQL, scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Flask, Docker, Git, Tableau, A/B testing

EXPERIENCE

Data Analyst Intern, Northwind Analytics Pty Ltd

02/2023 - 11/2023

Worked within the analytics team supporting internal reporting and a customer churn project.

- Built a churn prediction model using scikit-learn and engineered features from transaction history, which the retention team adopted into their weekly workflow.
- Automated a manual Excel reporting process into a Python script using Pandas, saving the team significant time each week.
- Created Tableau dashboards used by the marketing team to track campaign performance.
- Wrote SQL queries against the company's PostgreSQL warehouse to extract customer cohorts for analysis.
- Presented findings from the churn model to the head of retention, who used the results to prioritise outreach to at-risk customers.

Research Assistant (Casual), Mallee University Data Science Lab

06/2022 - 01/2023

Assisted a senior researcher with a project classifying satellite imagery of crop health.

- Labelled and pre-processed a dataset of several thousand satellite image tiles for a crop classification model.
- Implemented a convolutional neural network in TensorFlow to classify crop health from the labelled imagery, which improved the research group's baseline model.
- Wrote documentation and a short internal report summarising the approach for future students joining the lab.

Barista and Shift Supervisor, Hazel & Co Cafe

03/2019 - 05/2022

Worked across front-of-house and supervisory duties while completing undergraduate study.

- Opened and closed the store, managed cash reconciliation, and trained new baristas on espresso technique and customer service standards.
- Built and maintained relationships with regular customers, many of whom specifically asked for me by name when ordering.
- Managed rostering for a small team of four to six staff across weekday and weekend shifts, balancing availability with peak trading periods.
- Handled customer complaints and resolved disputes over orders and billing, often diffusing frustrated customers before they escalated to the manager.
- Took stock of inventory weekly, placed orders with suppliers for milk, coffee beans and pastries, and helped redesign the layout of the display counter to make it easier for customers to see the pastry options.
- Covered shifts for colleagues during busy periods including Christmas and university exam breaks, and was occasionally asked to help train staff at a second location owned by the same franchise.
- Participated in monthly staff meetings where rostering, stock levels and customer feedback were discussed, and contributed suggestions around opening hours during exam periods.

Retail Assistant, Bright Sports Store

11/2017 - 02/2019

Part time retail role during the final years of high school and the first year of undergraduate study.

- Assisted customers with selecting footwear and sporting equipment, operated the point-of-sale system, and processed returns and exchanges.
- Merchandised new stock according to the store's seasonal planograms and helped with the end-of-financial-year stocktake.
- Opened the store on weekend mornings and was trusted with the till float and security procedures.

PROJECTS

Movie Recommendation Engine

GitHub: github.com/dforsythe/movie-recommender

Built a collaborative filtering recommendation system in Python using the MovieLens dataset, implementing both matrix factorisation and a neighbourhood-based approach, then compared them using RMSE on a held-out test set. Wrapped the final model in a small Flask API with a simple front end for querying recommendations by user ID.

Road Sign Image Classifier

GitHub: github.com/dforsythe/sign-classifier

Trained a convolutional neural network in PyTorch to classify images of road signs into their respective categories, using data augmentation to address class imbalance in the training set. Containerised the inference pipeline with Docker so it could be run without manually installing dependencies.

Retail Demand Forecasting (Capstone Project)

GitHub: github.com/dforsythe/demand-forecasting

Capstone project for the Master of Data Science, forecasting weekly product demand for a simulated retail dataset using gradient boosting models compared against a baseline seasonal naive approach. Presented the project to classmates and faculty as part of the final capstone showcase.

EDUCATION

Master of Data Science, Mallee University

02/2021 - 12/2022

Core coursework: Statistical Machine Learning, Deep Learning, Big Data Systems, Data Visualisation, Applied Natural Language Processing, Time Series Analysis, Research Methods in Data Science, Cloud Computing for Data Science, Ethics in Artificial Intelligence, Database Systems, Optimisation Methods, Capstone Project. Completed individual assignments in every core course, including a semester-long group assignment in Big Data Systems building a Spark pipeline over a synthetic retail dataset, a series of weekly lab exercises in Deep Learning covering feedforward networks, CNNs and RNNs, and a take-home exam in Statistical Machine Learning covering regularisation, cross-validation and ensemble methods. Also completed a reflective journal assignment in Ethics in Artificial Intelligence discussing bias in facial recognition systems, and a final written report in Research Methods summarising a mini literature review on explainable AI techniques.

Bachelor of Commerce (Economics major), Mallee University

02/2017 - 11/2020

Completed a standard Bachelor of Commerce degree with a major in Economics and a minor in Statistics. Relevant coursework included Microeconomics, Macroeconomics, Introductory Statistics, Business Analytics, Marketing Principles and Organisational Behaviour. Completed a final-year research project analysing the relationship between minimum wage changes and regional youth unemployment using publicly available government datasets.

CERTIFICATES

- DeepLearning.AI TensorFlow Developer Professional Certificate, 2022
- Google Data Analytics Professional Certificate, 2021

HOBBIES AND INTERESTS

In my spare time I enjoy rock climbing at the local indoor gym, where I climb most weekends with a small group of friends from university. I also enjoy cooking, particularly experimenting with recipes from different cuisines, and I keep a small vegetable garden at home. I am an avid reader of both fiction and popular science books, and I try to read at least one book a month. I also enjoy playing social basketball on weekends and occasionally volunteer at a local community garden project on weekends when my schedule allows.

REFEREES

Available on request.

Additional referee details:

Dr. Susan Whitfield, Senior Lecturer, Mallee University Data Science Lab, available via university email on request.

Mark Chen, Store Manager, Hazel & Co Cafe, available via phone on request.

Priya Nair, Analytics Manager, Northwind Analytics Pty Ltd, available via email on request.